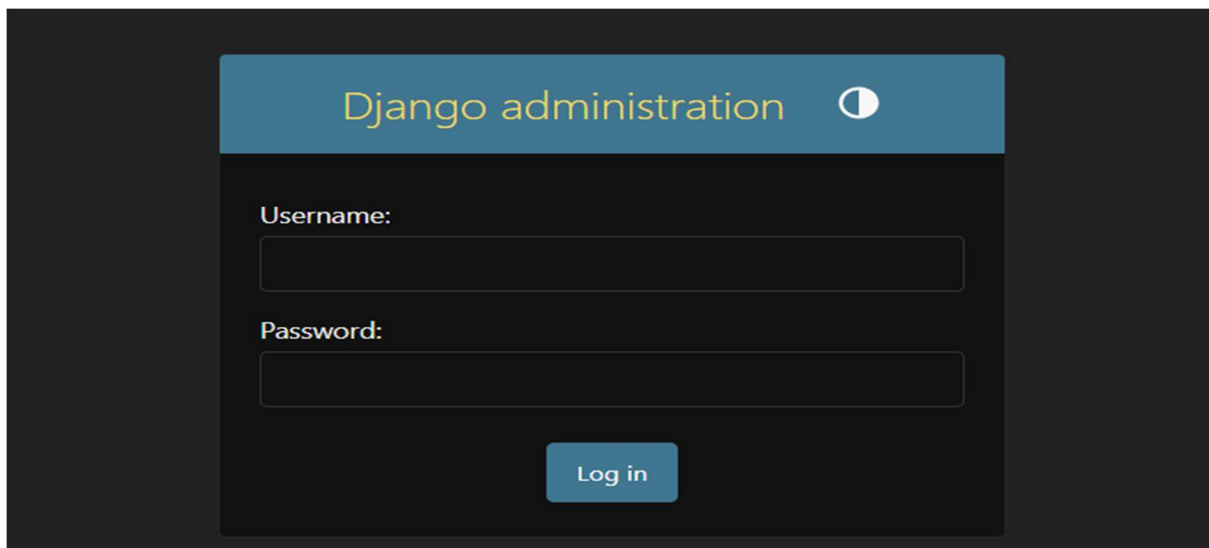
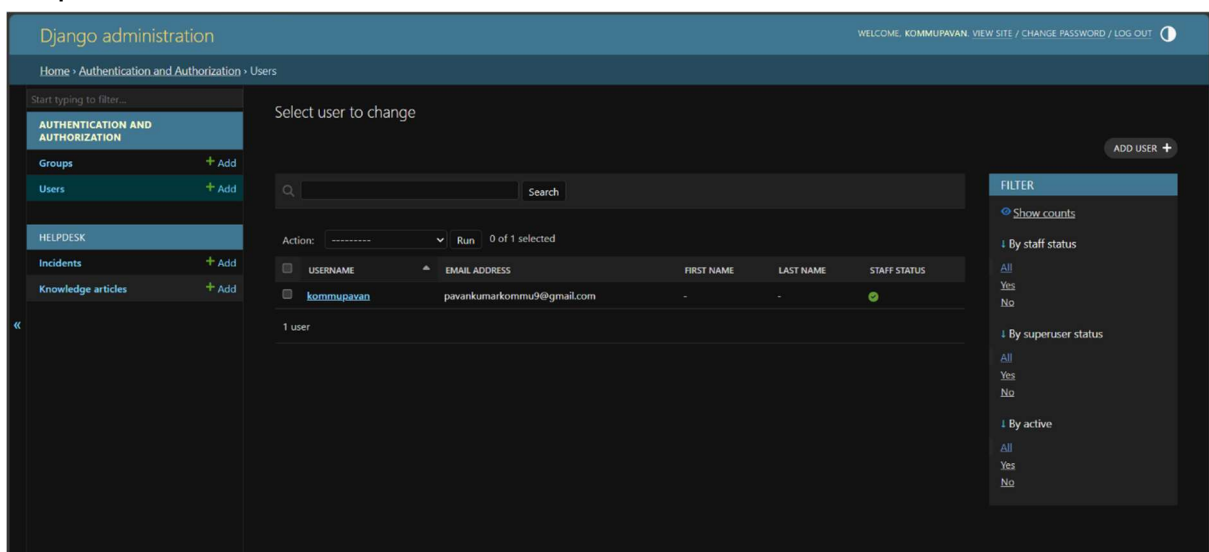


## Django Dashboard

The **Django Dashboard** is the administrative control panel of the **GenAI-Powered IT Service Desk Assistant**. It provides a secure and centralized platform for administrators to manage incidents, maintain the knowledge base, monitor support data, and control the overall functioning of the application. Built using Django's powerful administration framework, the dashboard enables efficient management of IT service operations without requiring direct interaction with the database.

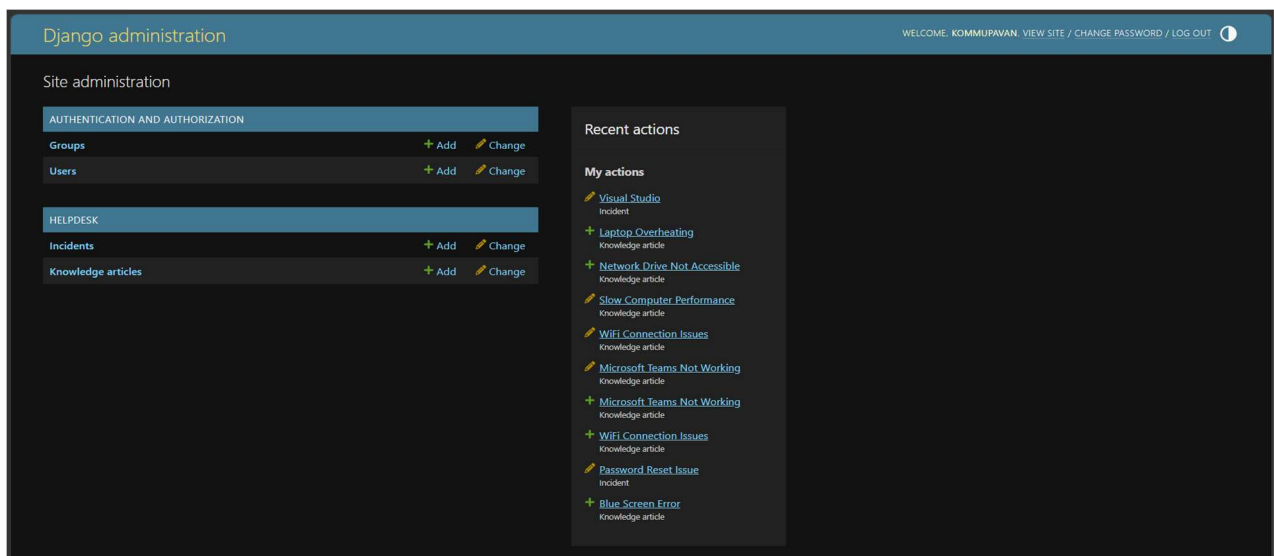


The dashboard is protected through Django's built-in authentication system, ensuring that only authorized administrators can access sensitive information and perform management tasks. This enhances the security and integrity of the application's data while simplifying administrative responsibilities.



## **Purpose of the Django Dashboard**

The primary purpose of the Django Dashboard is to provide an easy-to-use interface for managing the application's backend. Instead of writing SQL queries or editing database records manually, administrators can perform all operations through a graphical web interface. The dashboard serves as the backbone of the project by maintaining the information that is later utilized by the Local AI model (Phi-3 via Ollama) to generate intelligent troubleshooting recommendations.



## **Features of the Django Dashboard**

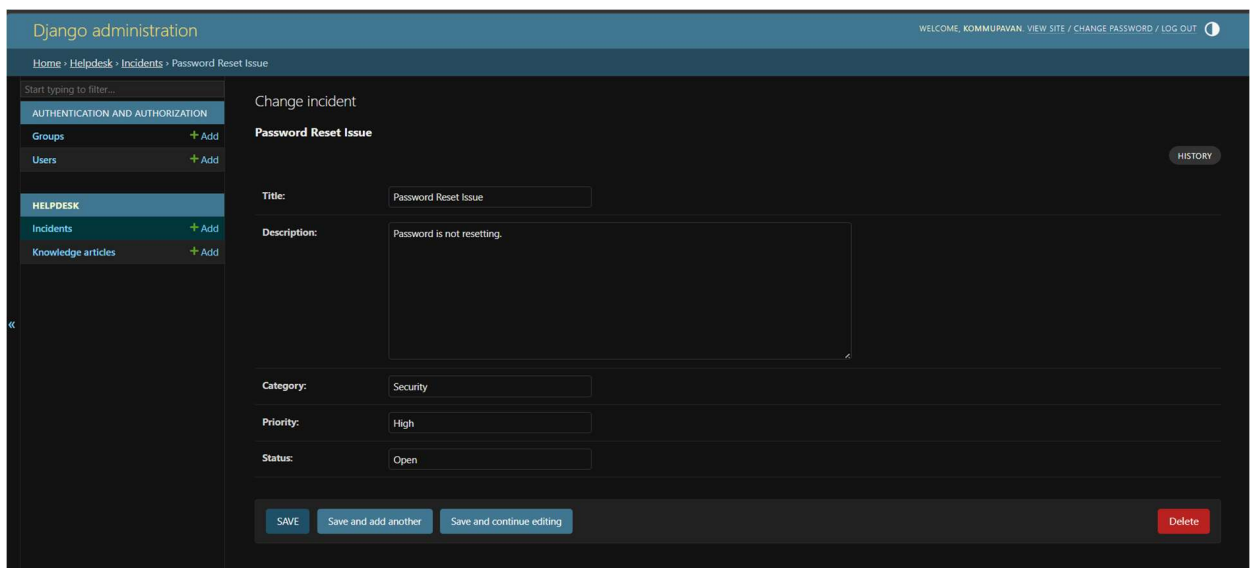
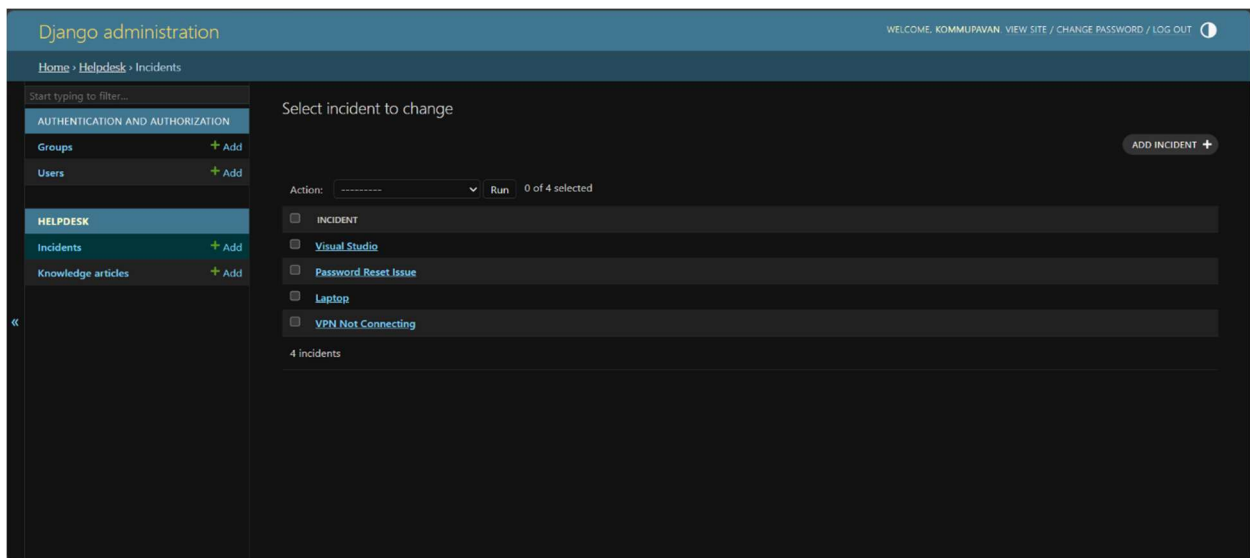
### **1. Incident Management**

The dashboard allows administrators to manage all IT incidents submitted by users through the application.

Administrators can:

- View all reported incidents
- Create new incidents
- Edit incident information
- Update incident status (Open, In Progress, Solved)
- Delete incorrect or duplicate incidents
- Monitor incident history

This functionality helps IT support teams efficiently organize and resolve support requests.



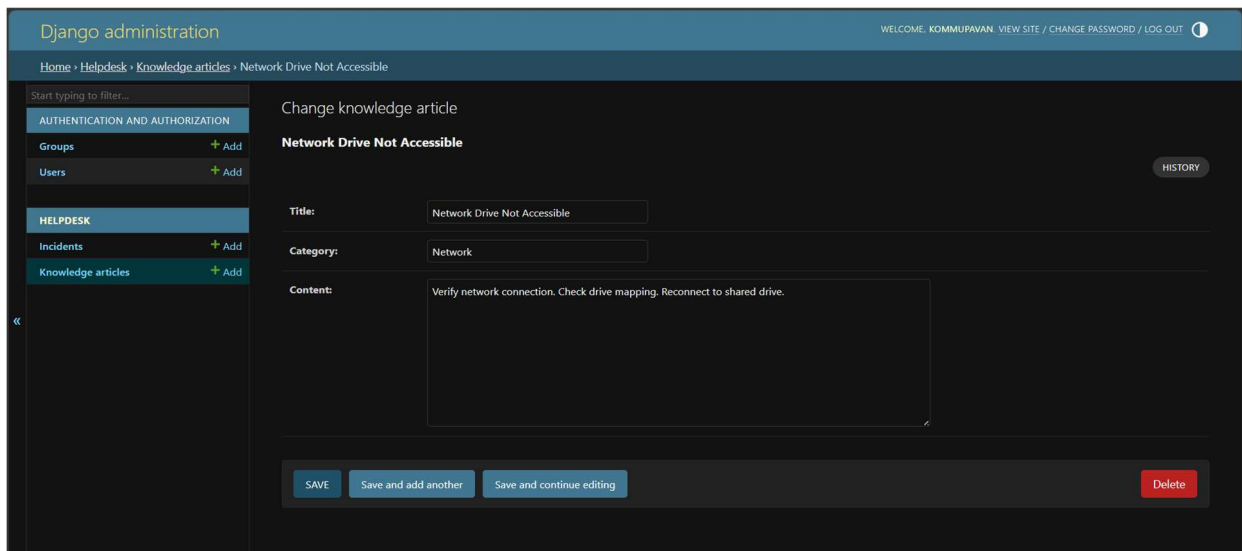
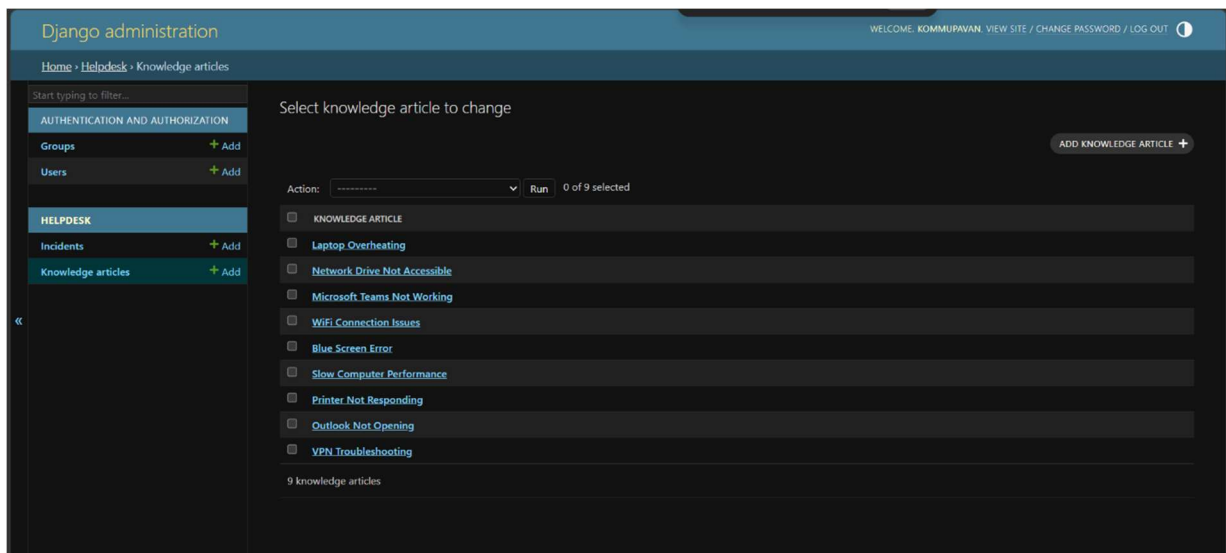
## 2. Knowledge Base Management

One of the core features of the project is the Knowledge Base, which stores solutions for common IT problems.

Through the Django Dashboard, administrators can:

- Add new troubleshooting articles
- Edit existing solutions
- Delete outdated articles
- Organize articles by category
- Maintain standardized troubleshooting procedures

The Knowledge Base acts as the first source of information whenever users ask questions in the AI Assistant.



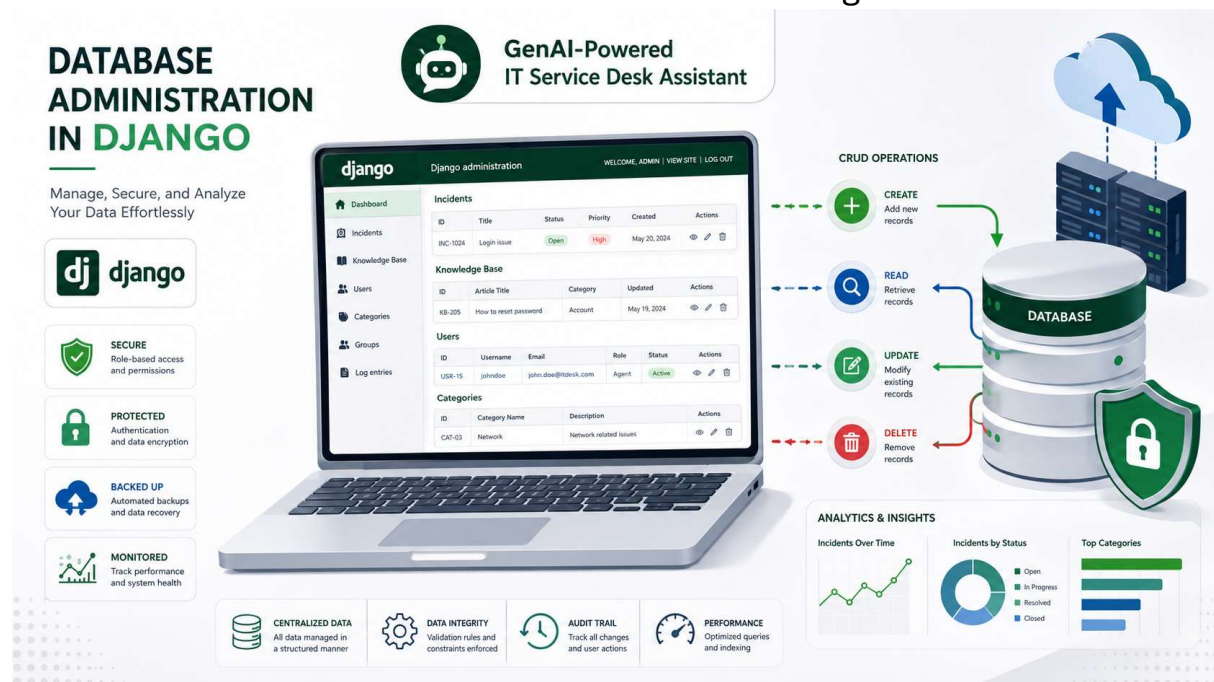
### 3. Database Administration

The dashboard provides complete access to the application's database records.

Administrators can:

- View all stored data
- Modify records instantly
- Remove unnecessary entries
- Maintain data consistency
- Perform CRUD (Create, Read, Update, Delete) operations

This eliminates the need for manual database management.



#### 4. User Authentication and Security

The Django Dashboard includes a secure authentication mechanism.

Security features include:

- Administrator login
- Password protection
- Session management
- CSRF protection
- Permission-based access

Only authenticated administrators can access the dashboard, protecting sensitive organizational data from unauthorized users.



## 5. Real-Time Data Management

Any modifications made through the dashboard are immediately reflected throughout the application.

For example:

- Adding a new Knowledge Base article makes it instantly searchable.
- Updating an incident status is immediately visible on the Incident Dashboard.
- Editing troubleshooting content updates the AI Assistant's knowledge source.

This ensures the system always operates using the latest available information.

## 6. Integration with the AI Assistant

The Django Dashboard plays a significant role in supporting the project's AI functionality.

When a user submits a query:

1. The AI Assistant first searches the Knowledge Base managed through the Django Dashboard.
2. If a relevant solution exists, it is displayed to the user.
3. The query is then forwarded to the Local AI model (Phi-3 running on Ollama).
4. The AI generates additional troubleshooting recommendations based on the user's issue.

This hybrid approach combines structured organizational knowledge with AI-generated insights, resulting in more accurate and comprehensive support.





## **7. Efficient Data Organization**

The dashboard organizes information into separate management modules such as:

- Incidents
- Knowledge Articles
- Categories
- Status Information

This structured organization improves usability and simplifies maintenance for administrators.

## **8. Easy Maintenance**

The Django Dashboard enables administrators to update the application without changing the source code.

Administrators can:

- Add new IT support solutions
- Modify troubleshooting procedures
- Remove obsolete content
- Keep the Knowledge Base current

This reduces maintenance effort and allows the application to evolve with organizational requirements.

## **Advantages of Using the Django Dashboard**

- Provides a secure administrative interface.
- Simplifies database management through a graphical interface.
- Supports efficient incident tracking and monitoring.
- Enables quick updates to the Knowledge Base.
- Reduces manual database operations.
- Automatically reflects updates in the user interface.
- Integrates seamlessly with the Local AI Assistant.
- Improves overall system maintainability and reliability.

## **Role of the Django Dashboard in the Project**

The Django Dashboard serves as the central management system of the GenAI-Powered IT Service Desk Assistant. It enables administrators to efficiently manage incidents and maintain the Knowledge Base that powers the application's intelligent support features. By integrating with the Local AI model (Phi-3 via Ollama), the dashboard ensures that users receive both organization-approved troubleshooting solutions and AI-generated recommendations. This combination enhances the quality, speed, and accuracy of IT support services while providing a secure, scalable, and easy-to-maintain administrative environment.